7.1: Simplifying and Verifying Trigonometric Identities

Recall the following trigonometric identities:

- Pythagorean Identities:

$$\begin{aligned} \cos^2(\theta) + \sin^2(\theta) &= 1 \\ 1 + \tan^2(\theta) &= \sec^2(\theta) \\ \cot^2(\theta) + 1 &= \csc^2(\theta) \end{aligned}$$

- Even-odd Identities

Even	Odd
$\cos(-\theta) = \cos(\theta)$	$\sin(-\theta) = -\sin(\theta)$
$\sec(-\theta) = \sec(\theta)$	$\csc(-\theta) = -\csc(\theta)$
	$\tan(-\theta) = -\tan(\theta)$
	$\cot(-\theta) = -\cot(\theta)$

- Reciprocal identities:

$$\sin(\theta) = \frac{1}{\csc(\theta)} \qquad \cos(\theta) = \frac{1}{\sec(\theta)} \qquad \tan(\theta) = \frac{1}{\cot(\theta)}$$

$$\csc(\theta) = \frac{1}{\sin(\theta)} \qquad \sec(\theta) = \frac{1}{\cos(\theta)} \qquad \cot(\theta) = \frac{1}{\tan(\theta)}$$

- Quotient identities:

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)} \qquad \cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$$

Example. Verify $\tan(\theta)\cos(\theta) = \sin(\theta)$.

Verify $\csc(\theta)\cos(\theta)\tan(\theta) = 1$.

Verify $(1+\sin(x))(1+\sin(-x)) = \cos^2(x)$

Verify $\dfrac{\sec^2(\theta)-1}{\sec^2(\theta)} = \sin^2(\theta)$

Example. Create an identity for the expression $2\tan(\theta)\sec(\theta)$ by rewriting strictly in terms of $\sin(\theta)$

Example. Solve the following:

$$2\cos^2(\theta) + \cos(\theta) - 1 = 0$$

Example. Factor the trigonometric expression $25 - 9\sin^2(\theta)$